OS ORANGE-2 ASSIGNMENT

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GitHub link: https://github.com/aniketrj45/PES2UG24CS063-pes-vcs.git

Phase 1 — Object Storage (Screenshots 1A & 1B)

Screenshot 1A — Should show all 3 tests passing.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ ./pes init
./test_objects
Initialized empty PES repository in .pes/
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Screenshot 1B — Shows sharded directories like `.pes/objects/d5/8213f...`

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Phase 2 — Tree Objects (Screenshots 2A & 2B)

Screenshot 2A — Both tree tests pass.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ make clean && make all
./pes init
./test_tree
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Initialized empty PES repository in .pes/
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Screenshot 2B — Raw binary tree format.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ find .pes/objects -type f
# copy one path, then:
xxd .pes/objects/XX/YYYY... | head -20
xxd: .pes/objects/XX/YYYY...: No such file or directory
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Phase 3 — Index / Staging (Screenshots 3A & 3B)

Screenshot 3A — Shows staged files, nothing unstaged, lots of untracked.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ make clean && make all
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_objects
  untracked:  Makefile
  untracked:  index.c
  untracked:  index.h
  untracked:  pes.h
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  test_tree
  untracked:  test_sequence.sh
  untracked:  README.md
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  object.c
  untracked:  commit.c
  untracked:  test_objects.c
  untracked:  tree.h
  untracked:  commit.h

Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Screenshot 3B — Two lines with mode, hash, mtime, size, filename.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776450947 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776450947 6 file2.txt
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Phase 4 — Commits & Log (Screenshots 4A, 4B, 4C)

Screenshot 4A — Three commits with hash, author, timestamp, message.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ ./pes init
echo "Hello" > hello.txt && ./pes add hello.txt && ./pes commit -m "Initial commit"
echo "World" >> hello.txt && ./pes add hello.txt && ./pes commit -m "Add world"
echo "Goodbye" > bye.txt && ./pes add bye.txt && ./pes commit -m "Add farewell"
./pes log
Initialized empty PES repository in .pes/
Committed: d94f0e9a7f5e... Initial commit
Committed: 86a3f0dcca1f... Add world
Committed: 72dbe535873e... Add farewell
commit 72dbe535873e7ad68f56a8f1bac85bcc699564281e7b30e87652269145dcaf0e
Author: AniketRJ <PES2UG24CS063>
Date:   1776451254

    Add farewell

commit 86a3f0dcca1fccb7fb21e00eab647edbbabfcd8cc50d8f4b4357f69c6de6fb41
Author: AniketRJ <PES2UG24CS063>
Date:   1776451254

    Add world

commit d94f0e9a7f5ed3de32cd35c04e1943d870fd47262c01691b84c0e9c04eb7da62
Author: AniketRJ <PES2UG24CS063>
Date:   1776451254

    Initial commit

Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Screenshot 4B — Many object files showing history growth.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: c5bdafeaa541... Initial commit

Log after first commit:
commit c5bdafeaa541b9c3f1cf6afde612310e38c015c577ca4cca06825638fdb10263
Author: AniketRJ <PES2UG24CS063>
Date:   1776451341

    Initial commit


--- Second Commit ---
Committed: dd298a47e7f6... Update file.txt

--- Third Commit ---
Committed: a20855508317... Add farewell

--- Full History ---
commit a208555083177ec8090da8ced5d86ef50d9463b580b1ada2321f894e91806bb6
Author: AniketRJ <PES2UG24CS063>
Date:   1776451341
```

//continuation screenshot

```
--- Full History ---
commit a208555083177ec8090da8ced5d86ef50d9463b580b1ada2321f894e91806bb6
Author: AniketRJ <PES2UG24CS063>
Date:   1776451341

    Add farewell

commit dd298a47e7f6e6fcb12950e81222c49e80860b1c73e1e553dfebc4bf8ec08b73
Author: AniketRJ <PES2UG24CS063>
Date:   1776451341

    Update file.txt

commit c5bdafeaa541b9c3f1cf6afde612310e38c015c577ca4cca06825638fdb10263
Author: AniketRJ <PES2UG24CS063>
Date:   1776451341

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
a208555083177ec8090da8ced5d86ef50d9463b580b1ada2321f894e91806bb6

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/a2/08555083177ec8090da8ced5d86ef50d9463b580b1ada2321f894e91806bb6
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/c5/bdafeaa541b9c3f1cf6afde612310e38c015c577ca4cca06825638fdb10263
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/dd/298a47e7f6e6fcb12950e81222c49e80860b1c73e1e553dfebc4bf8ec08b73

=== All integration tests completed ===
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

Screenshot 4C — `HEAD` shows `ref: refs/heads/main`, main shows latest commit hash.

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
72dbe535873e7ad68f56a8f1bac85bcc699564281e7b30e87652269145dcaf0e
ref: refs/heads/main
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

<u>Final Integration Test</u>

```
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: a1a0bbc25473... Initial commit

Log after first commit:
commit a1a0bbc25473b37376f29a9a1d70a3a46d0d75bda83ede51f0c8006b1e57f588
Author: AniketRJ <PES2UG24CS063>
Date:   1776451617

    Initial commit


--- Second Commit ---
Committed: 83bd06e99819... Update file.txt

--- Third Commit ---
Committed: 7cdf21d05a30... Add farewell

--- Full History ---
commit 7cdf21d05a3009b2fef3df360ea48e49d35402d45205c0fca85bb1641c84f5ba
Author: AniketRJ <PES2UG24CS063>
Date:   1776451617
```

//continuation screenshot

```
--- Full History ---
commit 7cdf21d05a3009b2fef3df360ea48e49d35402d45205c0fca85bb1641c84f5ba
Author: AniketRJ <PES2UG24CS063>
Date:   1776451617

    Add farewell

commit 83bd06e9981930ddc06208d6dc50494852b3b761dc18f430eade61749b9238b6
Author: AniketRJ <PES2UG24CS063>
Date:   1776451617

    Update file.txt

commit a1a0bbc25473b37376f29a9a1d70a3a46d0d75bda83ede51f0c8006b1e57f588
Author: AniketRJ <PES2UG24CS063>
Date:   1776451617

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
7cdf21d05a3009b2fef3df360ea48e49d35402d45205c0fca85bb1641c84f5ba

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/7c/df21d05a3009b2fef3df360ea48e49d35402d45205c0fca85bb1641c84f5ba
.pes/objects/83/bd06e9981930ddc06208d6dc50494852b3b761dc18f430eade61749b9238b6
.pes/objects/a1/a0bbc25473b37376f29a9a1d70a3a46d0d75bda83ede51f0c8006b1e57f588
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
Aniketrj@ubuntu-24-04-lts:~/PES2UG24CS063-pes-vcs$
```

<u>Phase 5 and 6: Analysis question answers</u>

Q 5.1

Update .pes pointers by verifying the branch file exists at .pes/refs/heads/, reading the target commit hash, and rewriting .pes/HEAD to ref: refs/heads/ using an atomic temp + rename.

Resolve the target snapshot by reading the commit object, its root tree, and recursively expanding tree objects to build the full path → blob mapping.

Update the working directory by writing each blob to its file with correct mode, removing

tracked files not present in the target tree, and recreating directories as needed. Rebuild the index by setting .pes/index entries to match the target tree with updated stat metadata.
This is complex because it must safely handle deletions, nested directories, permissions, partial failures, and avoid data loss when uncommitted changes overlap with target changes.

Q 5.2

Use a 3-way comparison per tracked file with base as the current HEAD tree blob hash, staged intent as the hash in .pes/index, and working file as stat plus recomputed blob hash.
Build the target tree mapping and for each path differing between current HEAD and target, check if the working directory is dirty relative to the current committed state.
If the file is dirty and the target also changes that path, refuse checkout. Also detect untracked file overwrite cases where an untracked working file would be overwritten by the target tree and refuse.
This enforces the rule that checkout should not overwrite local non-committed content.

Q 5.3

In detached HEAD, .pes/HEAD stores a commit hash directly instead of a branch ref, and new commits still work by pointing to the previous detached commit while HEAD moves forward. The risk is that no branch references these commits, making them potentially unreachable.
Recovery involves creating a branch ref pointing to that commit chain or switching to a branch after anchoring the detached commit. If not recovered and no refs point to them, future GC may delete them.

Q 6.1

Use a mark-and-sweep approach where the mark phase initializes a hash set, seeds it with all branch tips, and performs DFS or BFS traversal marking commits, trees, and blobs while enqueueing referenced objects. The sweep phase scans all files in .pes/objects and deletes those whose hashes are not in the reachable set, optionally cleaning empty directories.
A hash set ensures O(1) expected membership checks and deduplication during traversal. In scale, traversal visits unique reachable objects, typically hundreds of thousands to a few

million, making the complexity proportional to reachable objects plus total objects on disk.

Q 6.2

A race occurs when commit writes blob and tree objects, then commit object, and finally updates the branch ref while GC runs before the ref update and cannot see the new chain. GC treats new objects as unreachable and deletes them, after which the branch points to a broken history with missing objects.

Real systems avoid this through synchronization and locking around ref updates and GC, reachability protections like reflogs and grace periods, avoiding immediate pruning of recent objects, and ensuring objects are durable before references move.